

Saeid Safari Abrazi

Embedded Systems & FPGA/SoC Engineer | IEC 61850 | IEEE 1588 PTP

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Tehran, Iran | Open to relocation and international opportunities

Embedded systems and FPGA/SoC engineer with 7+ years of experience developing real-time industrial and power-system products. Strong hands-on background in Xilinx Zynq, C/C++, Verilog, hardware-software integration, Ethernet communication, IEC 61850 GOOSE/Sampled Values, and precision synchronization using IEEE 1588 PTP, GPS, IRIG-B, and SNTP. Experienced in cross-functional technical leadership from architecture and prototyping through integration and product readiness.

CORE EXPERTISE

FPGA & SoC	Xilinx Zynq-7000, Verilog/VHDL, custom IP, AXI4/AXI4-Lite, BRAM, DDR, timing, ILA, hardware-software partitioning
Embedded	C/C++, bare-metal, FreeRTOS, embedded Linux/PetaLinux, drivers, interrupts, boot/flash, board bring-up, debugging
Protocols & Timing	IEC 61850 GOOSE/SV, IEEE 1588 PTP, Ethernet, TCP/UDP, LwIP, VLAN, SNTP/NTP, GPS/PPS, IRIG-B, Modbus
Electronics & Tools	Altium, ADC/DAC, SPI/I2C/UART/CAN/USB, Vivado/SDK, C#/.NET, Python, MATLAB, Git, Linux, Wireshark

PROFESSIONAL EXPERIENCE

Cross-Functional Team Leader & Technical Product Manager Mar 2022 - Present
Vebko Amirkabir Tehran, Iran

- Coordinate embedded, FPGA, electronics, desktop software, testing, and product activities for relay-testing and grid-automation products.
- Translate customer and product requirements into system architecture, technical priorities, development plans, and validation criteria.
- Lead integration of communication, precision timing, waveform generation, hardware interfaces, and engineering software across multidisciplinary teams.
- Support technical decision-making, troubleshooting, risk resolution, and the transition of R&D; concepts toward dependable product releases.

R&D; Electronic / Embedded & FPGA Engineer Nov 2019 - Present
Vebko Amirkabir Tehran, Iran

- Develop embedded firmware and FPGA subsystems on Xilinx Zynq-7000 using C/C++, Verilog, custom IP, AXI interfaces, BRAM, DDR, and bare-metal software.
- Implement IEC 61850 Sampled Values and GOOSE publisher/subscriber functionality integrated with real-time device control and signal-processing paths.
- Design and debug IEEE 1588 PTP synchronization, including utility-oriented profiles, peer delay, clock selection, correction handling, and holdover behavior.
- Integrate GPS/PPS, IRIG-B, SNTP/NTP, Ethernet PHYs, ADCs, flash, USB, and other board-level interfaces; perform bring-up and packet-level troubleshooting.
- Develop deterministic FPGA waveform and data-transfer architectures for synchronized multi-channel test equipment.

Electronic Technician

Early-career technical experience

2015 - 2017
Iran

- Built practical experience in electronic assembly, measurement, troubleshooting, repair, and hardware-focused technical work.

SELECTED ENGINEERING PROJECTS

IEEE 1588 PTP Time Synchronization for Power Utility Devices

C/C++ | Zynq | Ethernet | P2P | Power Utility Profiles

Designed and implemented precision clock synchronization functions for a Zynq-based industrial device. The work covered PTP message processing, peer-delay measurement, correction fields, best-master selection, clock state handling, offset/delay filtering, and integration with GPS, IRIG-B, and the internal FPGA timebase.

IEC 61850 GOOSE and Sampled Values Communication

IEC 61850 | GOOSE | SV | VLAN | Real-time integration

Developed publisher and subscriber functionality for event and sampled-waveform exchange, including packet parsing, dataset handling, VLAN-aware communication, logical processing, and integration with device test and protection functions.

Deterministic Multi-Channel FPGA Waveform Engine

Verilog | AXI | BRAM | DDR | Interpolation | Real-time timing

Built programmable-logic and firmware paths for synchronized waveform generation, memory transfer, interpolation, channel processing, and tightly timed output control in a relay-test platform.

Next-Generation Relay Test and Grid Automation Platforms

FPGA/SoC | Embedded | Electronics | Product integration

Contributed to product architecture and development for test-device generations combining analog/digital I/O, real-time signal generation, communication protocols, precision timing, custom electronics, and Windows engineering software.

GPS, IRIG-B, and Network Time Integration

GPS/PPS | IRIG-B | SNTP/NTP | Embedded timing

Implemented and integrated multiple external time sources, including GPS UTC/PPS, IRIG-B decoding, and network time services, with consistent application to the device and FPGA timing domains.

EDUCATION

Bachelor of Science - Aviation Electronics

B.Sc.

Multidisciplinary electronics education covering avionics, embedded systems, communication, instrumentation, and hardware-oriented engineering.

ADDITIONAL STRENGTHS

Technical leadership	Cross-functional planning, system architecture, debugging leadership, technical communication, and product-focused decision-making.
Development approach	Hands-on, measurement-driven engineering with strong attention to deterministic behavior, reliability, maintainability, and system-level integration.
Languages	Persian (native), English (professional working proficiency), German (beginner).